

Web Application Lab



Name : Anuska Nath

Roll : 002311001003

Dept : Information Technology

Year : UG-3

Section : A1

(Teacher incharge)

(Date)

Assignment - 1

Q1. Write a program to display a message when a button is clicked.

Use:

- #HTML button
- #JavaScript function
- #alert() OR text change using DOM

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Button Click Message</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 80px;
    }
    button {
      padding: 8px 16px;
      font-size: 14px;
    }
  </style>
</head>
<body>
  <button onclick="showMessage()">Click Me</button>
  <p id="message"></p>
  <script>
    function showMessage() {
      document.getElementById("message").innerHTML = "Button was clicked!";
    }
  </script>
</body>
</html>
```



Q2. Write a program to change the text of a paragraph when a button is clicked.

Use:

```
#getElementById()  
#innerHTML
```

Code:

```
<!DOCTYPE html>  
<html>  
<head>  
  <title>Change Paragraph Text</title>  
  <style>  
    body {  
      font-family: Arial;  
      text-align: center;  
      margin-top: 80px;  
    }  
    button {  
      padding: 6px 12px;  
    }  
  </style>  
</head>  
<body>  
  <p id="para">This is the original paragraph text.</p>  
  <button onclick="changeText()">Click Me</button>  
  <script>  
    function changeText() {  
      document.getElementById("para").innerHTML =  
        "The paragraph text has been changed!";  
    }  
  </script>  
</body>  
</html>
```

This is the original paragraph text.

Click Me

The paragraph text has been changed!

Click Me

Q3. Write a program to add two numbers using JavaScript and display the result.

Use:

#Two input boxes

#Button

#Result display using DOM

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Add Two Numbers</title>
</head>
<body>
  <input type="number" id="num1" placeholder="Enter first number" />
  <input type="number" id="num2" placeholder="Enter second number" />
  <br /><br />
  <button onclick="addNumbers()">Add</button>
  <p id="result"></p>
  <script>
    function addNumbers() {
      var a = document.getElementById("num1").value;
      var b = document.getElementById("num2").value;
      var sum = Number(a) + Number(b);
      document.getElementById("result").innerHTML = "Result: " + sum;
    }
  </script>
</body>
</html>
```

Enter first number	Enter second number
Add	

5	7
Add	
Result: 12	

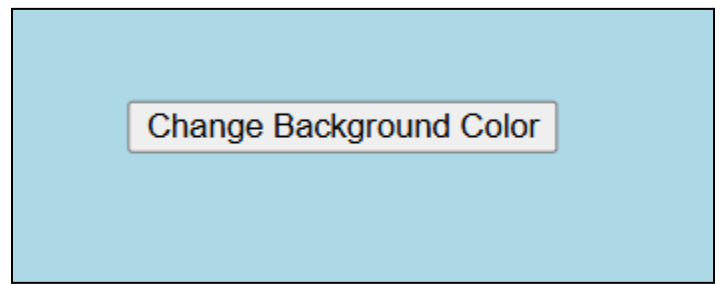
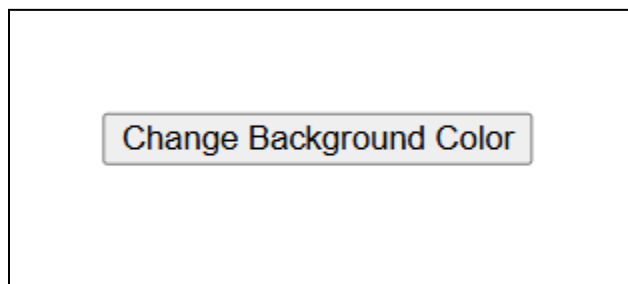
Q4. Write a program to change the background color of the page on button click.

Use:

```
#document.body.style.backgroundColor
```

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Change Background Color</title>
  <style>
    button{
      position: absolute;
      top: 50%;
      left: 50%;
      transform: translate(-50%, -50%);
    }
  </style>
</head>
<body>
  <button onclick="changeColor()">Change Background Color</button>
  <script>
    function changeColor() {
      document.body.style.backgroundColor = "lightblue";
    }
  </script>
</body>
</html>
```



Q5. Write a program to display the current date and time on button click.

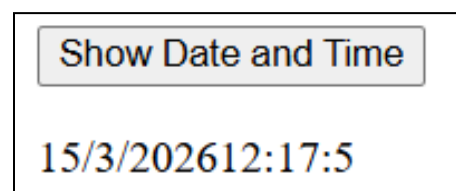
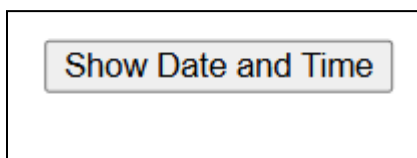
Use:

#Date() object

#DOM to show output

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Display Date and Time</title>
</head>
<body>
  <button onclick="showDateTime()">Show Date and Time</button>
  <p id="datetime"></p>
  <script>
    function showDateTime() {
      var now = new Date();
      const hour = now.getHours();
      const minute = now.getMinutes();
      const second = now.getSeconds();
      const Day = now.getDate();
      const Month = now.getMonth() + 1;
      const Year = now.getFullYear();
      const str1 = `${Day}/${Month}/${Year}`;
      const str2 = `${hour}:${minute}:${second}`;
      document.getElementById("datetime").innerHTML = str1 + str2;
    }
  </script>
</body>
</html>
```



Assignment - 2

Q1. Write a program to take a number from an input box and check whether the number is even or odd. Display the result on the webpage without using alert().

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Q1</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 80px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
<body>
  <input type="number" id="num1" placeholder="Enter a number here" />
  <button onclick="check()">Check</button>
  <p id="para"></p>
  <script>
    function check() {
      const num1 = document.getElementById("num1").value;
      let result = num1 % 2 == 0 ? "Even" : "Odd";
      document.getElementById("para").innerHTML = `The Number is ${result}`;
    }
  </script>
</body>
</html>
```

The Number is Even

The Number is Odd

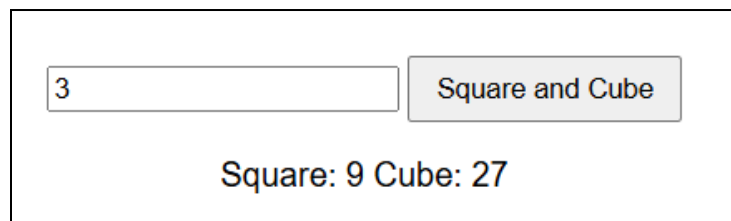
Q2. Write a program to take a number from the user and display its square and cube when a button is clicked.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Q2</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 80px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
<body>
  <input type="number" id="num1" placeholder="Enter a number here" />
  <button onclick="check()">Square and Cube</button>
  <p id="para"></p>
  <script>
    function check() {
      const num1 = document.getElementById("num1").value;
      let square = num1 * num1;
      let cube = square * num1;
      document.getElementById(
        "para"
      ).innerHTML = `Square: ${square} Cube: ${cube}`;
    }
  </script>
</body>
</html>
```



The image shows the initial state of the web application. It features a text input field with the placeholder text "Enter a number here" and a button labeled "Square and Cube". Both are centered within a light gray rectangular container.



The image shows the state of the web application after the button has been clicked with the value 3. The input field now contains the number "3". Below the input field and button, the text "Square: 9 Cube: 27" is displayed, indicating the calculated results.

Q3. Write a program to count how many times a button is clicked and display the count dynamically on the webpage.

Code:

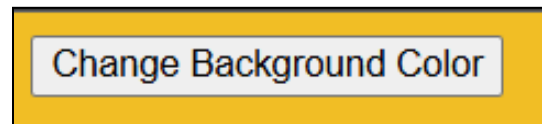
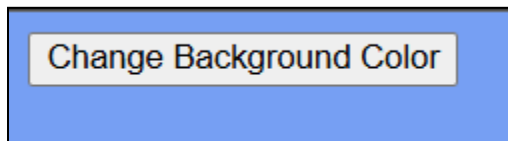
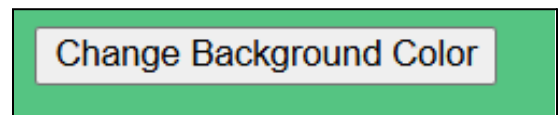
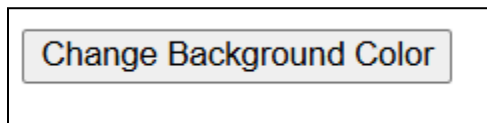
```
<!DOCTYPE html>
<html>
<head>
  <title>Q3</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 80px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
<body>
  <button onclick="increaseCount()">Press</button>
  <p id="para">button press count: 0</p>
  <script>
    let count = 0;
    function increaseCount() {
      count++;
      const str = `button press count: ${count}`;
      document.getElementById("para").innerHTML = str;
    }
  </script>
</body>
</html>
```



Q4. Write a program to change the background color of the page randomly every time a button is clicked.

Code:

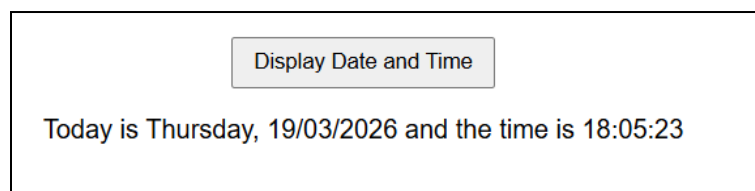
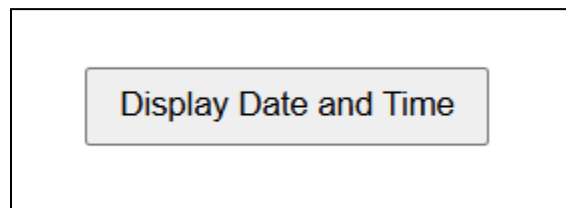
```
<!DOCTYPE html>
<html>
<head>
  <title>Q4</title>
</head>
<body>
  <button onclick="changeColor()">Change Background Color</button>
  <script>
    function getRandomColor() {
      var letters = "0123456789ABCDEF";
      var color = "#";
      for (var i = 0; i < 6; i++) {
        color += letters[Math.floor(Math.random() * 16)];
      }
      return color;
    }
    function changeColor() {
      document.body.style.backgroundColor = getRandomColor();
    }
  </script>
</body>
</html>
```



Q5. Write a program to display the current date, time, and day (Monday, Tuesday, etc.) on button click.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Q5</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
<body>
  <button onclick="displayDateTime()">Display Date and Time</button>
  <p id="dateTime"></p>
  <script>
    function displayDateTime() {
      var currentDate = new Date();
      var day = currentDate.toLocaleString('en-us', {
        weekday: 'long'
      });
      var date = currentDate.toLocaleDateString();
      var time = currentDate.toLocaleTimeString();
      document.getElementById("dateTime").innerText = "Today is " + day + ", " +
date + " and the time is " + time;
    }
  </script>
</body>
</html>
```

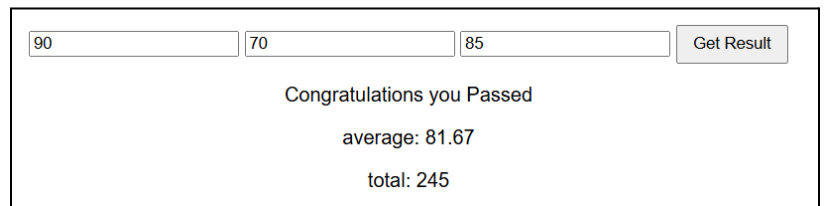


Q6. Write a program to take marks of three subjects from input boxes and display the total marks, average, and result (Pass/Fail).

```
<!DOCTYPE html>
<html>
<head>
  <title>Q6</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
```



Subject 1: Marks Subject 2: Marks Subject 3: Marks Get Result



90 70 85 Get Result

Congratulations you Passed

average: 81.67

total: 245

```
<body>
  <input type="number" name="sub1" id="sub1" placeholder="Subject 1: Marks">
  <input type="number" name="sub2" id="sub2" placeholder="Subject 2: Marks">
  <input type="number" name="sub3" id="sub3" placeholder="Subject 3: Marks">
  <button onclick="getResult()">Get Result</button>
  <p id="result"></p>
  <p id="avg"></p>
  <p id="total"></p>
  <script>
    function getResult() {
      const m1 = parseInt(document.getElementById('sub1').value);
      const m2 = parseInt(document.getElementById('sub2').value);
      const m3 = parseInt(document.getElementById('sub3').value);
      const total = m1 + m2 + m3;
      const avg = total / 3;
      const result = (m1 >= 40 && m2 >= 40 && m3 >= 40) ? 'Congratulations you
Passed' : 'Better luck next time';
      document.getElementById('result').innerHTML = result;
      document.getElementById('avg').innerHTML = "average: " + avg.toFixed(2);
      document.getElementById('total').innerHTML = "total: " + total;
    }
  </script>
</body>
</html>
```

Q7. Write a program to show or hide a paragraph when a button is clicked (toggle functionality).

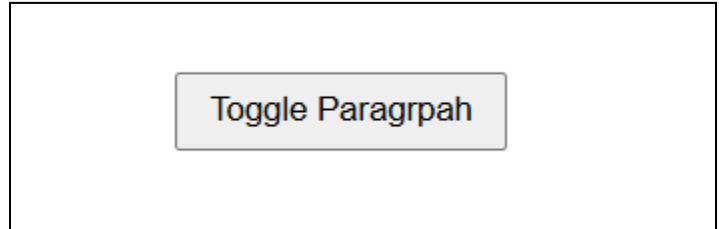
Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Q7</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
```

```
<body>
  <button onclick="toggleView()">Toggle Paragrpah</button>
  <p id="para" style="display: none">
```

Lorem, ipsum dolor sit amet consectetur adipisicing elit. Porro consequuntur officiis neque magnam dolorum exercitationem ea soluta dicta ipsa odio temporibus velit perspiciatis iusto, nesciunt animi ab magni consectetur tempora. Iusto debitis nesciunt dolorum voluptate. Minima, dicta! Fugit sequi similique esse ipsam maiores non illo natus facilis ducimus saepe asperiores quam perspiciatis labore atque nemo, accusantium error rem, veniam repellat. Blanditiis reiciendis quam illo similique a fugit saepe impedit nihil, quo atque vero autem et accusamus at, debitis ab tenetur veniam, quis maxime id beatae modi mollitia natus? Maiores repellendus et, illo molestias ea eos reprehenderit? Id dolore beatae eos.

```
</p>
<script>
  const para = document.getElementById("para");
  function toggleView() {
    para.style.display = para.style.display === "block" ? "none" : "block";
  }
</script>
</body>
</html>
```



Toggle Paragrpah

Lorem, ipsum dolor sit amet consectetur adipisicing elit. Porro consequuntur officiis neque magnam dolorum exercitationem ea soluta dicta ipsa odio temporibus velit perspiciatis iusto, nesciunt animi ab magni consectetur tempora. Iusto debitis nesciunt dolorum voluptate. Minima, dicta! Fugit sequi similique esse ipsam maiores non illo natus facilis ducimus saepe asperiores quam perspiciatis labore atque nemo, accusantium error rem, veniam repellat. Blanditiis reiciendis quam illo similique a fugit saepe impedit nihil, quo atque vero autem et accusamus at, debitis ab tenetur veniam, quis maxime id beatae modi mollitia natus? Maiores repellendus et, illo molestias ea eos reprehenderit? Id dolore beatae eos.

Q8. Write a program to validate a text box input and display a warning message if the input field is left empty.

```
<!DOCTYPE html>
<html>
<head>
  <title>Q8</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
<body>
  <input type="text" id="textBox" placeholder="Enter something" />
  <button onclick="validateInput()">Validate Input</button>
  <p id="warningMessage" style="color: red"></p>
  <script>
    function validateInput() {
      var input = document.getElementById("textBox").value;
      if (input === "") {
        document.getElementById("warningMessage").innerText =
          "Please fill out this field.";
      } else {
        document.getElementById("warningMessage").innerText = "Valid!";
      }
    }
  </script>
</body>
</html>
```



A rectangular container with a light gray border. Inside, on the left, is a text input field with a light gray border and the placeholder text "Enter something". To the right of the input field is a button with a light gray background and the text "Validate Input".



A rectangular container with a light gray border. Inside, on the left, is a text input field with a light gray border containing the text "abc". To the right of the input field is a button with a light gray background and the text "Validate Input". Below the input field and button, centered, is the text "Valid!" in red.



A rectangular container with a light gray border. Inside, on the left, is a text input field with a light gray border and the placeholder text "Enter something". To the right of the input field is a button with a light gray background and the text "Validate Input". Below the input field and button, centered, is the text "Please fill out this field." in red.

Q9. Write a program to display the largest number among three numbers entered by the user.

```
<!DOCTYPE html>
<html>
<head>
  <title>Q9</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 50px;
    }
    button {
      padding: 6px 12px;
    }
  </style>
</head>
<body>
  <input type="number" name="num1" id="num1" placeholder="Number 1">
  <input type="number" name="num2" id="num2" placeholder="Number 2">
  <input type="number" name="num3" id="num3" placeholder="Number 3">
  <button onclick="getResult()">Find Largest</button>
  <p id="result"></p>
  <script>
    function getResult() {
      const n1 = parseInt(document.getElementById('num1').value);
      const n2 = parseInt(document.getElementById('num2').value);
      const n3 = parseInt(document.getElementById('num3').value);
      const num = Math.max(n1, n2, n3);
      document.getElementById('result').innerHTML = "largest number: " + num;
    }
  </script>
</body>
</html>
```

Number 1	Number 2	Number 3	Find Largest
---------------------------------------	---------------------------------------	---------------------------------------	---

7	9	2	Find Largest
largest number: 9			

Assignment - 3

Q1. Write an HTML page including required JavaScript that:

- Takes a number from a text field
- Accepts numbers only in the range 0 to 999
- Displays the number in words
- Does not accept:
 - o Four or more digit numbers
 - o Alphabets
 - o Special characters

Validate

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Q1</title>
</head>
<body>
  <input type="text" id="num1" placeholder="Enter a number here" />
  <button onclick="check()">Validate</button>
  <p id="para"></p>
  <script>
    function check() {
      const num = document.getElementById("num1").value;
      const p = document.getElementById("para");
      for (let c = 0; c < num.length; c++) {
        if (isNaN(Number(num[c]))) {
          p.innerText = "INVALID";
          return;
        }
      }
      if (num.length >= 4) {
        p.innerText = "Invalid";
        return;
      }
      const num1 = parseInt(num);
      if (num1 < 0) {
        p.innerText = "Invalid";
        return;
      }
      console.log(numberToWords(num1));
      p.innerText = numberToWords(num1);
    }
  </script>
</body>
</html>
```

Validate

eight hundred ninety seven

Validate

INVALID

Validate

Invalid


```

function numberToWords(num) {
  if (num === 0) return 'zero';
  const units = ['', 'one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight', 'nine'];
  const teens = ['ten', 'eleven', 'twelve', 'thirteen', 'fourteen', 'fifteen', 'sixteen',
'seventeen', 'eighteen', 'nineteen'];
  const tens = ['', '', 'twenty', 'thirty', 'forty', 'fifty', 'sixty', 'seventy', 'eighty',
'ninety'];
  const scales = ['', 'thousand', 'million', 'billion', 'trillion'];
  function convertChunk(n) {
    let chunk = "";
    if (n >= 100) {
      chunk += units[Math.floor(n / 100)] + ' hundred ';
      n %= 100;
    }
    if (n < 10) {
      chunk += units[n];
    } else if (n >= 10 && n < 20) {
      chunk += teens[n - 10];
    } else {
      chunk += tens[Math.floor(n / 10)] + (n % 10 !== 0 ? ' ' + units[n % 10] : "");
    }
    return chunk.trim();
  }

  let words = "";
  let scaleIndex = 0;
  while (num > 0) {
    if (num % 1000 !== 0) {
      const chunkWords = convertChunk(num % 1000);
      words = chunkWords + (scales[scaleIndex] ? ' ' + scales[scaleIndex] : "")
+ (words ? ' ' + words : "");
    }
    num = Math.floor(num / 1000);
    scaleIndex++;
  }
  return words.trim();
}
</script>
</body>
</html>

```

Q2. Design the following static web pages required for an Online Book Store:

a) Home Page: Must contain three frames

b) Login Page

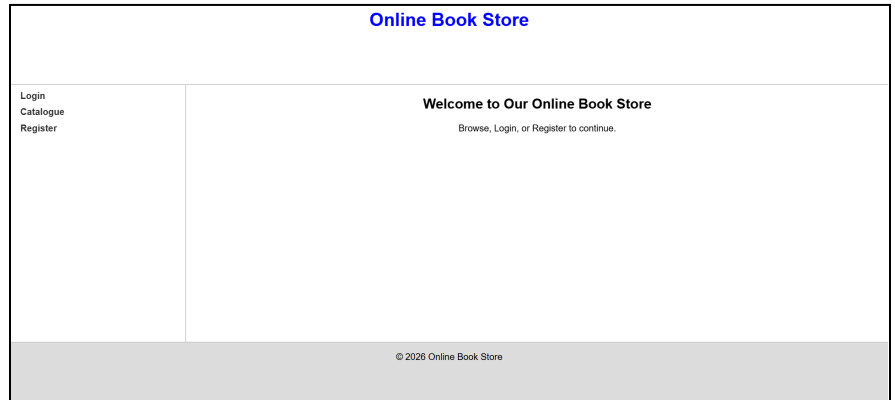
c) Catalogue Page

- Display details of all available books
- Use a table to show:
 - o Book Name
 - o Author
 - o Price

d) Registration Page

Q2.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Book Store - Home</title>
</head>
<frameset rows="20%,*,15%" border="2">
  <frame src="header.html" name="topFrame">
    <frameset cols="20%,*">
      <frame src="menu.html" name="leftFrame">
        <frame src="welcome.html" name="mainFrame">
      </frameset>
    <frame src="footer.html" name="bottomFrame">
  </frameset>
</html>
```



Login.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Login</title>
<style>
  body {
    font-family: Arial;
  }
  table {
    margin: auto;
  }
</style>
</head>
<body>
```

Login	
Username:	
Password:	
Login Clear	

```

<h2 align="center">Login</h2>
<form>
  <table cellpadding="8">
    <tr>
      <td>Username:</td>
      <td><input type="text" name="username" required></td>
    </tr>
    <tr>
      <td>Password:</td>
      <td><input type="password" name="password" required></td>
    </tr>
    <tr>
      <td colspan="2" align="center">
        <input type="submit" value="Login">
        <input type="reset" value="Clear">
      </td>
    </tr>
  </table>
</form>
</body>
</html>

```

Menu.html:

```

<html>
<head>
  <title>
    Menu</title>
  <style>
    body {
      font-family: Arial;
    }
    a {
      display: block;
      margin: 10px;
      text-decoration: none;
      color: #333;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <a href="login.html" target="mainFrame">Login</a>

```

Login
Catalogue
Register

```

    <a href="catalogue.html" target="mainFrame">Catalogue</a>
    <a href="register.html" target="mainFrame">Register</a>
</body>
</html>

```

Welcome.html:

```

<!DOCTYPE html>
<html>
<head>
    <title>Welcome</title>
</head>
<body style="font-family:Arial; text-align:center;">
    <h2>Welcome to Our Online Book Store</h2>
    <p>Browse, Login, or Register to continue.</p>
</body>
</html>

```

Welcome to Our Online Book Store

Browse, Login, or Register to continue.

Catalogue.html:

```

<!DOCTYPE html>
<html>
<head>
    <title>Catalogue</title>
    <style>
        body {
            font-family: Arial;
        }
        table {
            border-collapse: collapse;
            width: 80%;
            margin: auto;
        }
        th,
        td {
            border: 1px solid #333;
            padding: 8px;
            text-align: center;
        }
        th {
            background: #4CAF50;
            color: white;
        }
    </style>

```

Book Catalogue

Book Name	Author	Price (₹)
C and C++	Anuska Nath	350
HTML & CSS	Kaustav Mondal	550
question setting	Aritra Mondal	650
Python Crash Course	Anish Das	600

```

</head>
<body>
  <h2 align="center">Book Catalogue</h2>
  <table>
    <tr>
      <th>Book Name</th>
      <th>Author</th>
      <th>Price (₹)</th>
    </tr>
    <tr>
      <td>C and C++</td>
      <td>Anuska Nath</td>
      <td>350</td>
    </tr>
    <tr>
      <td>HTML & CSS</td>
      <td>Kaustav Mondal</td>
      <td>550</td>
    </tr>
    <tr>
      <td>question setting</td>
      <td>Aritra Mondal</td>
      <td>650</td>
    </tr>
    <tr>
      <td>Python Crash Course</td>
      <td>Anish Das</td>
      <td>600</td>
    </tr>
  </table>
</body>
</html>

```

Register.html:

```

<!DOCTYPE html>
<html>
<head>
  <title>Registration</title>
  <style>
    body {
      font-family: Arial;
    }

```

User Registration	
Full Name:	
Email:	
Username:	
Password:	
Confirm Password:	
Register Clear	

```

        table {
            margin: auto;
        }
    </style>
</head>
<body>
    <h2 align="center">User Registration</h2>
    <form>
        <table cellpadding="8">
            <tr>
                <td>Full Name:</td>
                <td><input type="text" required></td>
            </tr>
            <tr>
                <td>Email:</td>
                <td><input type="email" required></td>
            </tr>
            <tr>
                <td>Username:</td>
                <td><input type="text" required></td>
            </tr>
            <tr>
                <td>Password:</td>
                <td><input type="password" required></td>
            </tr>
            <tr>
                <td>Confirm Password:</td>
                <td><input type="password" required></td>
            </tr>
            <tr>
                <td colspan="2" align="center">
                    <input type="submit" value="Register">
                    <input type="reset" value="Clear">
                </td>
            </tr>
        </table>
    </form>
</body>
</html>

```

Header.html:

```
<html>
```

```

<head>
  <title>
    header
  </title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      color: blue;
    }
  </style>
</head>
<body>
  <h1>Online Book Store</h1>
</body>
</html>

```

Footer.html:

```

<!DOCTYPE html>
<html>
<head>
  <title>Footer</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      background: #ddd;
    }
  </style>
</head>
<body>
  <p>© 2026 Online Book Store</p>
</body>
</html>

```

Q3. Create a web page to display your Class Time Table using:

- **HTML table**
- **Proper headings for days and periods**

Code:

```

<!DOCTYPE html>
<html lang="en">

```

```

<head>
  <meta charset="UTF-8">
  <title>Class Time Table Question 3</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
    }
    h2,
    h3 {
      margin: 5px;
    }
    table {
      border-collapse: collapse;
      margin: 20px auto;
      width: 95%;
    }
    th,
    td {
      border: 1px solid black;
      padding: 8px;
      font-size: 14px;
    }
    th {
      background-color: #f2f2f2;
    }
    .break {
      writing-mode: vertical-rl;
      text-orientation: upright;
      font-weight: bold;
      background-color: #e6e6e6;
    }
  </style>
</head>
<body>
  <h2>BE IT 3<sup>rd</sup> Year</h2>
  <h3>2025–26 (2<sup>nd</sup> Semester)</h3>
  <h3>Jadavpur University</h3>
  <table>
    <tr>
      <th>Day / Time</th>
      <th>10:00–11:00</th>
    </tr>
  </table>

```


<th>11:00–12:00</th>
<th>12:00–1:00</th>
<th>1:00–2:00</th>
<th>2:00–2:30</th>
<th>2:30–3:30</th>
<th>3:30–4:30</th>
<th>4:30–5:30</th>

<th>Monday</th>	<td>Design & Analysis of Algorithm (SGC)</td>	<td colspan="3">Web Application Lab (A1) (GF) Multimedia Lab (A2) (MB, SGC)</td>
<td rowspan="5" class="break">BREAK</td>	<td colspan="3">Web Application Lab (A2) (GF) Multimedia Lab (A3) (MB, SGC)</td>	

<th>Tuesday</th>	<td></td>	<td colspan="2">Design & Analysis of Algorithm (SGC)</td>	<td>Information Security (MB)</td>	<td colspan="2">Web Technologies – II (GF)</td>	<td></td>
------------------	-----------	--	---------------------------------------	--	-----------

<th>Wednesday</th>	<td colspan="3">Web Application Lab (A3) (GF) Multimedia Lab (A1) (MB, SGC)</td>	<td>Multimedia Tutorial (MB, SGC)</td>	<td colspan="2">Elective-II: Big Data (PB)</td>	<td></td>
--------------------	---	---	--	-----------

<th>Thursday</th>	<td></td>	<td>Elective-II: Big Data (PB)</td>	<td colspan="2">Information Security (MB)</td>	<td>Web Technologies – II (GF)</td>
-------------------	-----------	--	---	--

```

        <td>Multimedia Coding & Communications<br>(GF)</td>
        <td></td>
    </tr>
    <tr>
        <th>Friday</th>
        <td></td>
        <td colspan="3">Big Data Lab (H)<br>(PB)</td>
        <td colspan="2">Multimedia Coding & Communications<br>(GF)</td>
        <td></td>
    </tr>
</table>
</body>
</html>

```

BE IT 3 rd Year								
2025–26 (2 nd Semester)								
Jadavpur University								
Day / Time	10:00–11:00	11:00–12:00	12:00–1:00	1:00–2:00	2:00–2:30	2:30–3:30	3:30–4:30	4:30–5:30
Monday	Design & Analysis of Algorithm (SGC)	Web Application Lab (A1) (GF) Multimedia Lab (A2) (MB, SGC)			B R E A K	Web Application Lab (A2) (GF) Multimedia Lab (A3) (MB, SGC)		
Tuesday		Design & Analysis of Algorithm (SGC)		Information Security (MB)		Web Technologies – II (GF)		
Wednesday	Web Application Lab (A3) (GF) Multimedia Lab (A1) (MB, SGC)			Multimedia Tutorial (MB, SGC)		Elective-II: Big Data (PB)		
Thursday		Elective-II: Big Data (PB)	Information Security (MB)			Web Technologies – II (GF)	Multimedia Coding & Communications (GF)	
Friday		Big Data Lab (H) (PB)				Multimedia Coding & Communications (GF)		

Q4. Design the required static web pages for an Online Shopping Cart application.

q4.html:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Online Shopping Cart</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #222;
            color: white;
        }
    </style>

```

```
header {
  background-color: #333;
  padding: 10px 0;
}
nav ul {
  list-style-type: none;
  text-align: center;
}
nav ul li {
  display: inline;
  margin: 0 15px;
}
nav ul li a {
  color: #fff;
  text-decoration: none;
}
main {
  padding: 20px;
  max-width: 1200px;
  margin: auto;
}

.product-grid {
  display: flex;
  justify-content: space-around;
}
.product-card {
  background-color: #fff;
  padding: 15px;
  margin: 10px;
  border-radius: 8px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
  text-align: center;
  width: 250px;
  color: black;
}
.product-card h3 {
  font-size: 18px;
  margin: 10px 0;
}
.product-card p {
  font-size: 16px;
```

```

        color: #555;
    }
    .product-card a {
        display: inline-block;
        margin-top: 10px;
        color: #333;
        text-decoration: none;
    }
    footer {
        background-color: #333;
        color: #fff;
        text-align: center;
        padding: 10px;
        position: fixed;
        width: 100%;
        bottom: 0;
    }
</style>
</head>
<body>
    <header>
        <nav>
            <ul>
                <li><a href="q4.html">Home</a></li>
                <li><a href="products.html">Products</a></li>
                <li><a href="cart.html">Cart</a></li>
                <li><a href="checkout.html">Checkout</a></li>
            </ul>
        </nav>
    </header>
    <main>
        <h1>Welcome to Our Online Store</h1>
        <section>
            <h2>Featured Products</h2>
            <div class="product-grid">
                <div class="product-card">
                    <h3>Product 1</h3>
                    <p>$50.00</p>
                    <a href="product-detail.html">View Details</a>
                </div>
                <div class="product-card">
                    <h3>Product 2</h3>

```

```

        <p>$30.00</p>
        <a href="product-detail.html">View Details</a>
    </div>
    <div class="product-card">
        <h3>Product 3</h3>
        <p>$70.00</p>
        <a href="product-detail.html">View Details</a>
    </div>
</div>
</section>
</main>
<footer>
    <p>&copy; 2026 Online Shopping Cart</p>
</footer>
</body>
</html>

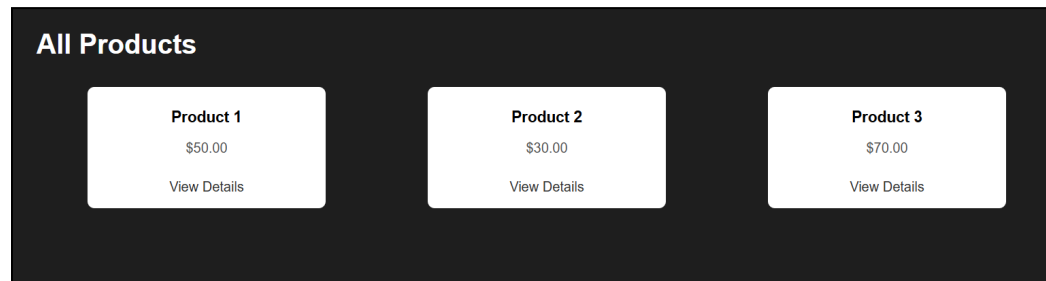
```

Products.html:

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Products - Online Shopping Cart</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #222;
            color: white;
        }
        header {
            background-color: #333;
            padding: 10px 0;
        }
        nav ul {
            list-style-type: none;
            text-align: center;
        }
        nav ul li {

```



```
    display: inline;
    margin: 0 15px;
}
nav ul li a {
    color: #fff;
    text-decoration: none;
}
main {
    padding: 20px;
    max-width: 1200px;
    margin: auto;
}
.product-grid {
    display: flex;
    justify-content: space-around;
}
.product-card {
    background-color: #fff;
    padding: 15px;
    margin: 10px;
    border-radius: 8px;
    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
    text-align: center;
    width: 250px;
    color: black;
}
.product-card h3 {
    font-size: 18px;
    margin: 10px 0;
}
.product-card p {
    font-size: 16px;
    color: #555;
}
.product-card a {
    display: inline-block;
    margin-top: 10px;
    color: #333;
    text-decoration: none;
}
footer {
    background-color: #333;
```

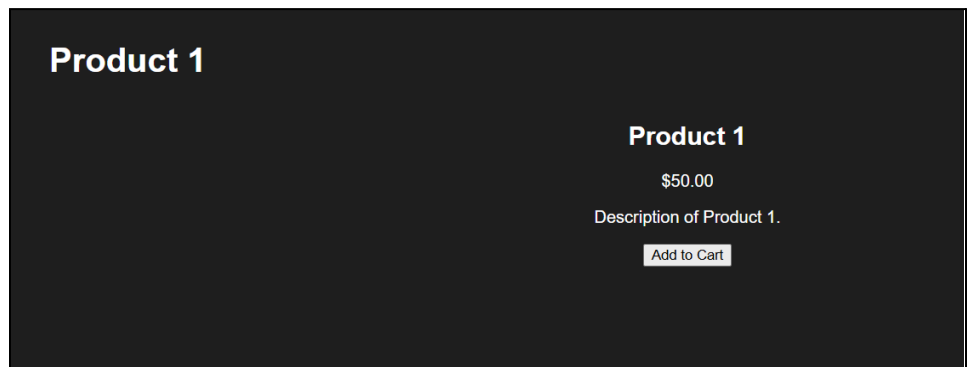
```
        color: #fff;
        text-align: center;
        padding: 10px;
        position: fixed;
        width: 100%;
        bottom: 0;
    }
</style>
</head>
<body>
    <header>
        <nav>
            <ul>
                <li><a href="q4.html">Home</a></li>
                <li><a href="products.html">Products</a></li>
                <li><a href="cart.html">Cart</a></li>
                <li><a href="checkout.html">Checkout</a></li>
            </ul>
        </nav>
    </header>
    <main>
        <h1>All Products</h1>
        <div class="product-grid">
            <div class="product-card">
                <h3>Product 1</h3>
                <p>$50.00</p>
                <a href="product-detail.html">View Details</a>
            </div>
            <div class="product-card">
                <h3>Product 2</h3>
                <p>$30.00</p>
                <a href="product-detail.html">View Details</a>
            </div>
            <div class="product-card">
                <h3>Product 3</h3>
                <p>$70.00</p>
                <a href="product-detail.html">View Details</a>
            </div>
        </div>
    </main>
    <footer>
        <p>&copy; 2026 Online Shopping Cart</p>
```



```

.product-detail {
  display: flex;
  justify-content: space-around;
  margin-top: 20px;
}
.product-info {
  text-align: center;
  width: 300px;
}
footer {
  background-color: #333;
  color: #fff;
  text-align: center;
  padding: 10px;
  position: fixed;
  width: 100%;
  bottom: 0;
}
</style>
</head>
<body>
  <header>
    <nav>
      <ul>
        <li><a href="q4.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart</a></li>
        <li><a href="checkout.html">Checkout</a></li>
      </ul>
    </nav>
  </header>
  <main>
    <h1>Product 1</h1>
    <div class="product-detail">
      <div class="product-info">
        <h2>Product 1</h2>
        <p>$50.00</p>
        <p>Description of Product 1.</p>
        <button>Add to Cart</button>
      </div>
    </div>
  </main>

```



```
<footer>
  <p>&copy; 2026 Online Shopping Cart</p>
</footer>
</body>
</html>
```

Cart.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Shopping Cart - Online Shopping Cart</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #222;
      color: white;
    }
    header {
      background-color: #333;
      padding: 10px 0;
    }
    nav ul {
      list-style-type: none;
      text-align: center;
    }
    nav ul li {
      display: inline;
      margin: 0 15px;
    }
    nav ul li a {
      color: #fff;
      text-decoration: none;
    }
    main {
      padding: 20px;
      max-width: 1200px;
      margin: auto;
    }
```

```

    }
    table {
        width: 100%;
        margin-top: 20px;
        border-collapse: collapse;
    }
    table th,
    table td {
        border: 1px solid #ddd;
        padding: 10px;
        text-align: center;
    }
    }
    footer {
        background-color: #333;
        color: #fff;
        text-align: center;
        padding: 10px;
        position: fixed;
        width: 100%;
        bottom: 0;
    }
}
</style>
</head>
<body>
<header>
<nav>
<ul>
    <li><a href="q4.html">Home</a></li>
    <li><a href="products.html">Products</a></li>
    <li><a href="cart.html">Cart</a></li>
    <li><a href="checkout.html">Checkout</a></li>
</ul>
</nav>
</header>
<main>
<h1>Your Shopping Cart</h1>
<table>
    <tr>
        <th>Product</th>
        <th>Price</th>
        <th>Quantity</th>
        <th>Total</th>
    </tr>
    <tr>
        <td>Product 1</td>
        <td>$50.00</td>
        <td>1</td>
        <td>$50.00</td>
    </tr>
    <tr>
        <td>Product 2</td>
        <td>$30.00</td>
        <td>2</td>
        <td>$60.00</td>
    </tr>
</table>
    <div>
        Total: $110.00
        <a href="#">Proceed to Checkout</a>
    </div>
</main>

```

Product	Price	Quantity	Total
Product 1	\$50.00	1	\$50.00
Product 2	\$30.00	2	\$60.00

Total: \$110.00

[Proceed to Checkout](#)

```

</tr>
<tr>
  <td>Product 1</td>
  <td>$50.00</td>
  <td><input type="number" value="1"></td>
  <td>$50.00</td>
</tr>
<tr>
  <td>Product 2</td>
  <td>$30.00</td>
  <td><input type="number" value="2"></td>
  <td>$60.00</td>
</tr>
</table>
<p>Total: $110.00</p>
<a href="checkout.html" style="color: #fff;">Proceed to Checkout</a>
</main>
<footer>
  <p>&copy; 2026 Online Shopping Cart</p>
</footer>
</body>
</html>

```

Checkout.html:

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Checkout - Online Shopping Cart</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 0;
      background-color: #222;
      color: white;
    }
    header {
      background-color: #333;
      padding: 10px 0;
    }

```

```

}
nav ul {
  list-style-type: none;
  text-align: center;
}
nav ul li {
  display: inline;
  margin: 0 15px;
}
nav ul li a {
  color: #fff;
  text-decoration: none;
}
main {
  padding: 20px;
  max-width: 1200px;
  margin: auto;
}
form {
  width: 400px;
  margin: auto;
  text-align: center;
}
form input,
form select,
form button {
  width: 100%;
  padding: 10px;
  margin: 10px 0;
  border-radius: 5px;
  border: 1px solid #ccc;
}
footer {
  background-color: #333;
  color: #fff;
  text-align: center;
  padding: 10px;
  position: fixed;
  width: 100%;
  bottom: 0;
}
</style>

```

Checkout

Name:

Address:

Payment Method:

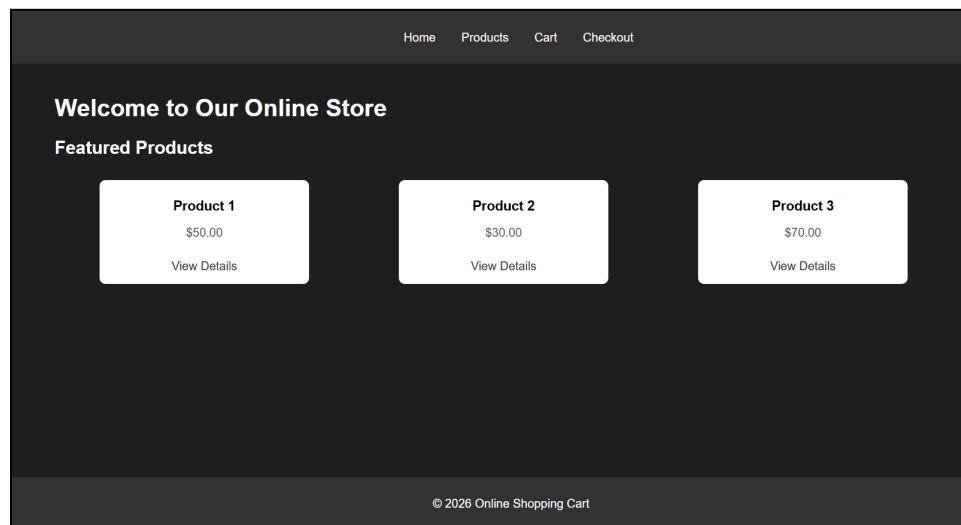
Credit Card
 

Complete Purchase

```

</head>
<body>
  <header>
    <nav>
      <ul>
        <li><a href="q4.html">Home</a></li>
        <li><a href="products.html">Products</a></li>
        <li><a href="cart.html">Cart</a></li>
        <li><a href="checkout.html">Checkout</a></li>
      </ul>
    </nav>
  </header>
  <main>
    <h1>Checkout</h1>
    <form action="#">
      <label for="name">Name:</label>
      <input type="text" id="name" name="name" required><br>
      <label for="address">Address:</label>
      <input type="text" id="address" name="address" required><br>
      <label for="payment">Payment Method:</label>
      <select id="payment" name="payment">
        <option value="credit-card">Credit Card</option>
        <option value="paypal">PayPal</option>
      </select><br>
      <button type="submit">Complete Purchase</button>
    </form>
  </main>
  <footer>
    <p>&copy; 2026 Online Shopping Cart</p>
  </footer>
</body>
</html>

```



Q5. Write JavaScript code to validate the following fields in a Registration Page:

- 1. First Name**
 - o Should contain only alphabets
 - o Minimum length: 6 characters
- 2. Password**
 - o Minimum length: 6 characters
- 3. E-mail ID**
 - o Must follow the standard pattern: name@domain.com
- 4. Mobile Number**
 - o Must contain exactly 10 digits
- 5. Last Name and Address**
 - o Should not be empty

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form Validation</title>
  <script>
    function validateForm() {
      var fname = document.getElementById("fname").value;
      var lname = document.getElementById("lname").value;
      var password = document.getElementById("password").value;
      var email = document.getElementById("email").value;
      var mobile = document.getElementById("mobile").value;
      var address = document.getElementById("address").value;
      var fnamePattern = /^[A-Za-z]{6,}$/;
      if (!fnamePattern.test(fname)) {
        alert("First Name must contain only alphabets and minimum 6 characters.");
        return false;
      }
      if (lname.trim() === "") {
        alert("Last Name cannot be empty.");
        return false;
      }
      if (password.length < 6) {
        alert("Password must be at least 6 characters long.");
        return false;
      }
      var emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
      if (!emailPattern.test(email)) {
```

```

        alert("Invalid Email ID format. Example: name@domain.com");
        return false;
    }
    var mobilePattern = /^[0-9]{10}$/;
    if (!mobilePattern.test(mobile)) {
        alert("Mobile Number must contain exactly 10 digits.");
        return false;
    }
    if (address.trim() === "") {
        alert("Address cannot be empty.");
        return false;
    }
    alert("Registration Successful!");
    return true;
}
</script>
</head>
<body>
    <h2>Registration Form</h2>
    <form onsubmit="return validateForm()">
        First Name:
        <input type="text" id="fname"><br><br>
        Last Name:
        <input type="text" id="lname"><br><br>
        Password:
        <input type="password" id="password"><br><br>
        Email ID:
        <input type="text" id="email"><br><br>
        Mobile Number:
        <input type="text" id="mobile"><br><br>
        Address:
        <textarea id="address"></textarea><br><br>
        <input type="submit" value="Register">
    </form>
</body>
</html>

```

Registration Form

First Name:

Last Name:

Password:

Email ID:

Mobile Number:

Address:

Q6. Design a web page using CSS that includes the following:

a) Use different font styles

b) Control image repetition using:

- background-repeat
- No-repeat

c) Define styles for links:

- a:link
- a:active
- a:hover
- A:visited

d) Add customized cursors for links

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>CSS Features Demo</title>
  <style>
    body {
      background-color: #2b2b2b;
      color: #f2f2f2;
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 30px;
    }
    /* Different font styles */
    h1 {
      font-family: Georgia, serif;
      text-align: center;
    }
    h2 {
      font-family: "Courier New", monospace;
      margin-top: 40px;
    }
    p {
      font-family: Verdana, sans-serif;
      font-size: 16px;
    }
    /* Background image - no repeat */
    .background-box {
      background-image: url("https://via.placeholder.com/800x200");
      background-repeat: no-repeat;
```

```

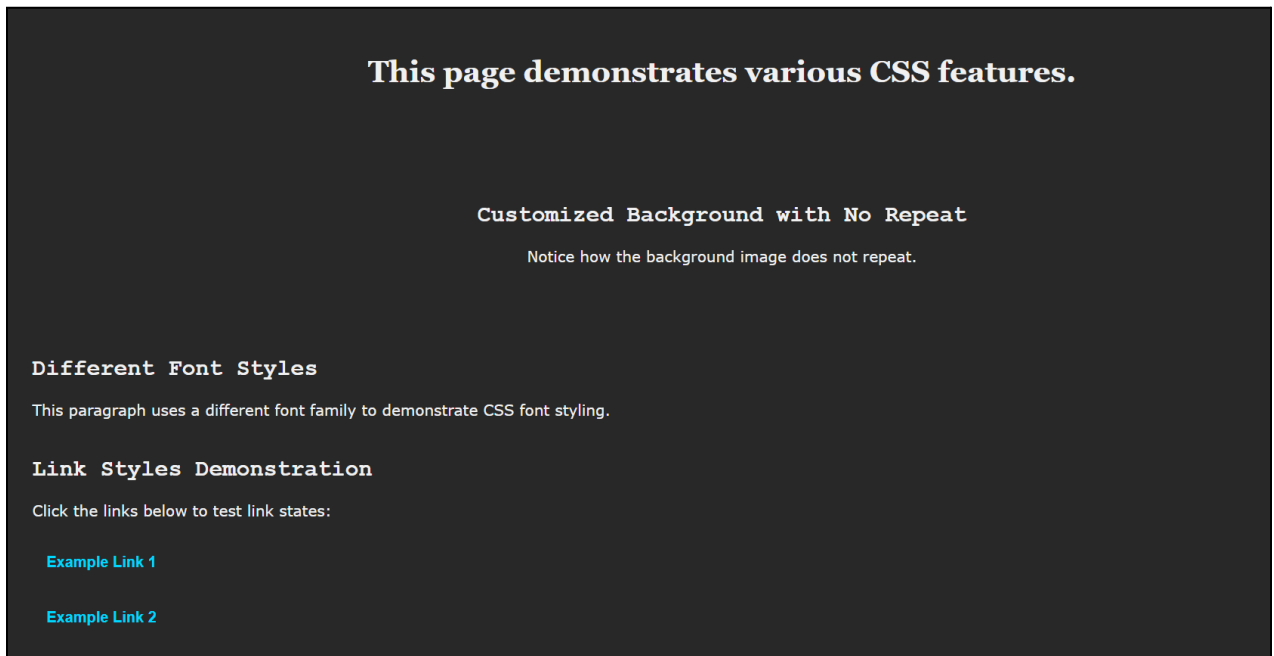
        background-size: cover;
        padding: 40px;
        margin: 40px 0;
        border-radius: 10px;
        text-align: center;
    }
    /* LINK STYLES */
    a {
        display: inline-block;
        margin: 10px 0;
        padding: 10px 15px;
        font-weight: bold;
    }
    /* unvisited link */
    a:link {
        color: #00d9ff;
        text-decoration: none;
    }
    /* visited link */
    a:visited {
        color: #b388ff;
    }
    /* mouse over link */
    a:hover {
        color: #ffffff;
        background-color: #00d9ff;
        border-radius: 5px;
        cursor: pointer;
    }
    /* active link */
    a:active {
        color: yellow;
        background-color: red;
    }
    /* custom cursor for links */
    a:hover {
        cursor: url("https://cur.cursors-4u.net/others/oth-1/oth50.cur"), pointer;
    }
</style>
</head>
<body>
    <h1>This page demonstrates various CSS features.</h1>

```

```

<div class="background-box">
  <h2>Customized Background with No Repeat</h2>
  <p>Notice how the background image does not repeat.</p>
</div>
<h2>Different Font Styles</h2>
<p>
  This paragraph uses a different font family to demonstrate CSS font styling.
</p>
<h2>Link Styles Demonstration</h2>
<p>Click the links below to test link states:</p>
<!-- LINKS YOU CAN SEE AND USE -->
<a href="https://example.com" target="_blank">Example Link 1</a><br>
<a href="https://example.com" target="_blank">Example Link 2</a>
</body>
</html>

```



Q7. Create a responsive web page for any one of the following:

- College Club
- Product Promotion Page
- Event Information Page

The page must contain:

1. Header
2. Navigation menu
3. Content section
4. Footer

Apply CSS3 features:

- Rounded corners
- Shadow effects
- Gradient background
- Transition effect
- Transform effect

(JavaScript may be used for button click / simple interaction)

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Tech Club</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <style>
    /* ===== Global Styles ===== */
    body {
      margin: 0;
      font-family: Arial, sans-serif;
      background: linear-gradient(to right, #1f4037, #99f2c8);
      /* Gradient background */
      color: #fff;
    }
    /* ===== Header ===== */
    header {
      text-align: center;
      padding: 30px;
      background: rgba(0, 0, 0, 0.3);
      box-shadow: 0 4px 10px rgba(0, 0, 0, 0.5);
      /* Shadow effect */
    }
    header h1 {
      margin: 0;
    }
    /* ===== Navigation ===== */
    nav {
      background: #111;
      text-align: center;
      padding: 10px 0;
    }
    nav a {
```

```
    color: white;
    text-decoration: none;
    margin: 0 15px;
    font-weight: bold;
    transition: color 0.3s ease;
    /* Transition effect */
}
nav a:hover {
    color: #99f2c8;
}
/* ===== Main Content ===== */
.container {
    max-width: 1000px;
    margin: 30px auto;
    padding: 20px;
}
.card {
    background: #ffffff;
    color: #333;
    padding: 20px;
    margin-bottom: 20px;
    border-radius: 15px;
    /* Rounded corners */
    box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);
    /* Shadow effect */
    transition: transform 0.3s ease;
    /* Transition */
}
.card:hover {
    transform: scale(1.03);
    /* Transform effect */
}
button {
    padding: 10px 20px;
    border: none;
    border-radius: 25px;
    background: linear-gradient(to right, #1f4037, #99f2c8);
    color: #fff;
    font-size: 16px;
    cursor: pointer;
    transition: transform 0.3s ease;
```

```

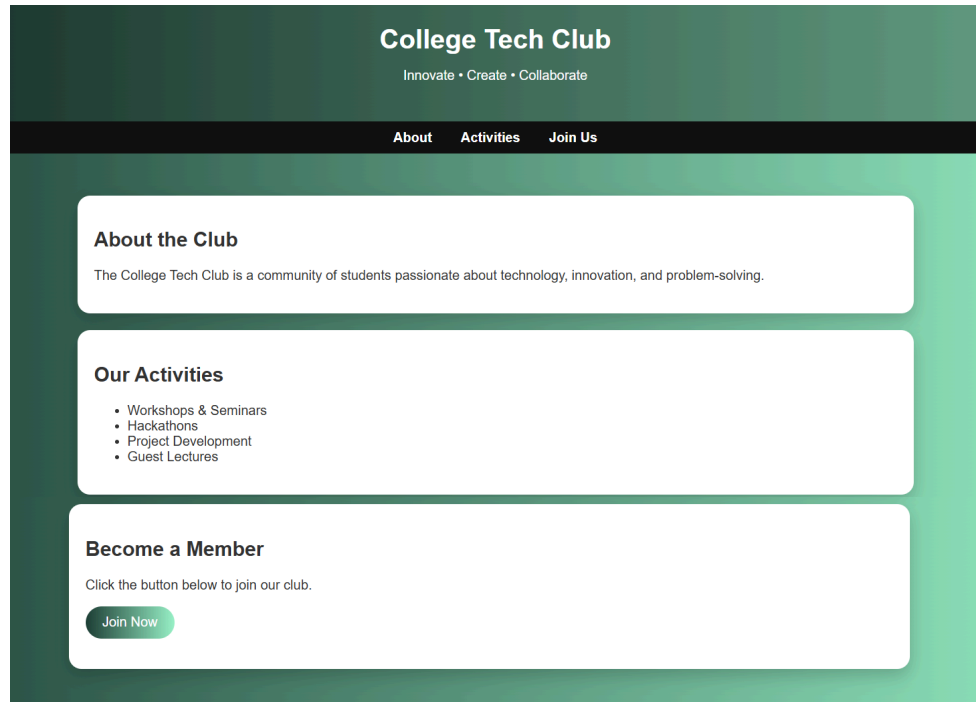
    }
    button:hover {
        transform: translateY(-3px);
        /* Transform effect */
    }
    /* ===== Footer ===== */
    footer {
        text-align: center;
        padding: 15px;
        background: #111;
        margin-top: 40px;
    }
    /* ===== Responsive Design ===== */
    @media (max-width: 600px) {
        nav a {
            display: block;
            margin: 10px 0;
        }
    }
</style>
</head>
<body>
    <!-- ===== Header ===== -->
    <header>
        <h1>College Tech Club</h1>
        <p>Innovate • Create • Collaborate</p>
    </header>
    <!-- ===== Navigation ===== -->
    <nav>
        <a href="#about">About</a>
        <a href="#activities">Activities</a>
        <a href="#join">Join Us</a>
    </nav>
    <!-- ===== Content Section ===== -->
    <div class="container">
        <div class="card" id="about">
            <h2>About the Club</h2>
            <p>
                The College Tech Club is a community of students passionate about
                technology, innovation, and problem-solving.
            </p>
        </div>
    </div>

```

```

<div class="card" id="activities">
  <h2>Our Activities</h2>
  <ul>
    <li>Workshops & Seminars</li>
    <li>Hackathons</li>
    <li>Project Development</li>
    <li>Guest Lectures</li>
  </ul>
</div>
<div class="card" id="join">
  <h2>Become a Member</h2>
  <p>Click the button below to join our club.</p>
  <button onclick="joinClub()">Join Now</button>
  <p id="message"></p>
</div>
</div>
<!-- ===== Footer ===== -->
<footer>
  <p>&copy; 2026 College Tech Club</p>
</footer>
<!-- ===== JavaScript ===== -->
<script>
  function joinClub() {
    document.getElementById("message").innerText =
      "Thank you for joining the College Tech Club!";
  }
</script>
</body>
</html>

```



Assignment - 3

1. Write a program to create a dynamic Student List Manager.

User can:

- Add student names
- Remove a student
- Display total number of students

Use:

- #DOM createElement()
- #appendChild()
- #Remove button per item

```
<!DOCTYPE html>
<html>
<head>
  <title>Student List Manager</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      padding: 20px;
    }
    input {
      margin: 5px;
      padding: 5px;
    }
    button {
      padding: 5px 10px;
      margin: 5px;
    }
    .student-item {
      margin: 8px 0;
      padding: 8px;
      background-color: #f4f4f4;
      border-radius: 5px;
    }
  </style>
</head>
<body>
  <h2>Student List Manager</h2>
  <input type="text" id="name" placeholder="Enter Name">
  <input type="number" id="age" placeholder="Enter Age">
  <input type="text" id="grade" placeholder="Enter Grade">
```

Student List Manager

Count of total students: 1

Name: Anuska | Age: 21 | Grade: A

Student List Manager

Count of total students: 2

Name: Anuska | Age: 21 | Grade: A

Name: Anwesha | Age: 27 | Grade: B


```

<button onclick="addStudent()">Add Student</button>
<hr>
<div id="count">
  <div>
    <span id="count">

      </span>
    </div>
  </div>
<div id="studentList"></div>
<script>
  let count = 0;

  function showTotal() {
    document.getElementById("count").textContent = "Count of total students: " +
count;
  }
  function addStudent() {
    count++;
    let name = document.getElementById("name").value;
    let age = document.getElementById("age").value;
    let grade = document.getElementById("grade").value;

    if (name === "" || age === "" || grade === "") {
      alert("Please fill all fields!");
      return;
    }
    let studentDiv = document.createElement("div");
    studentDiv.className = "student-item";
    let studentText = document.createElement("span");
    studentText.textContent = "Name: " + name +
      " | Age: " + age +
      " | Grade: " + grade;
    let deleteBtn = document.createElement("button");
    deleteBtn.textContent = "Delete";
    deleteBtn.style.marginLeft = "10px";

    deleteBtn.onclick = function() {
      count--;
      showTotal();
      studentDiv.remove();
    };
  };

```

```

        studentDiv.appendChild(studentText);
        studentDiv.appendChild(deleteBtn);
        document.getElementById("studentList").appendChild(studentDiv);
        showTotal();
        document.getElementById("name").value = "";
        document.getElementById("age").value = "";
        document.getElementById("grade").value = "";
    }
</script>
</body>
</html>

```

2. Write a program to generate a random OTP (6-digit) and verify it.

Requirements:

- Generate OTP on button click
- User enters OTP
- Show “Valid OTP” or “Invalid OTP”

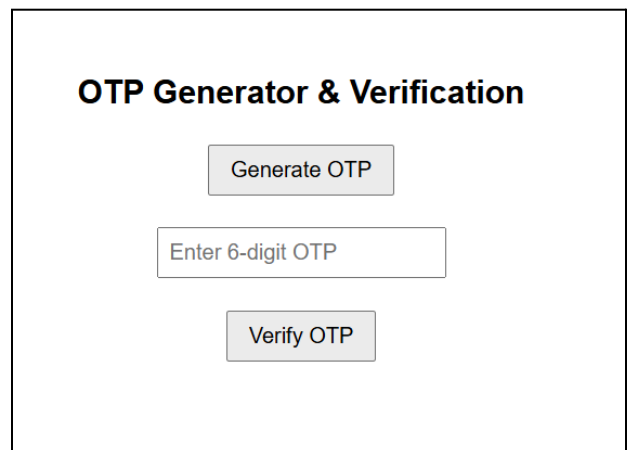
Use:

- **#Math.random()**
- **#DOM validation**

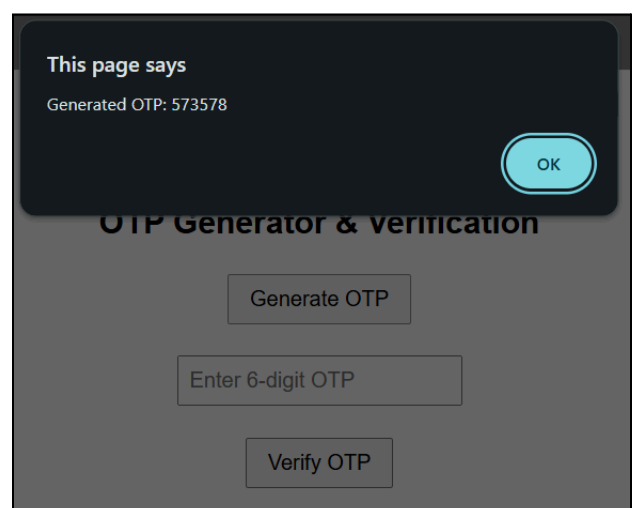
```

<!DOCTYPE html>
<html>
<head>
<title>OTP Generator & Validator</title>
<style>
    body {
        font-family: Arial, sans-serif;
        text-align: center;
        margin-top: 100px;
    }
    input {
        padding: 8px;
        font-size: 16px;
    }
    button {
        padding: 8px 15px;
        font-size: 16px;
        margin: 5px;
        cursor: pointer;
    }
    #message {

```



The screenshot shows a web application titled "OTP Generator & Verification". It features a "Generate OTP" button at the top. Below it is a text input field with the placeholder text "Enter 6-digit OTP". At the bottom is a "Verify OTP" button. The interface is clean and centered.



This screenshot shows the same web application after a successful OTP generation. A dark modal box at the top displays the message "This page says" followed by "Generated OTP: 573578" and an "OK" button. The background of the application is dimmed, but the "Generate OTP", "Enter 6-digit OTP", and "Verify OTP" elements are still visible.

```

        margin-top: 15px;
        font-weight: bold;
    }
</style>
</head>
<body>
    <h2>OTP Generator & Verification</h2>
    <button onclick="generateOTP()">Generate OTP</button>
    <br><br>
    <input type="text" id="userOTP" placeholder="Enter 6-digit OTP">
    <br><br>
    <button onclick="verifyOTP()">Verify OTP</button>
    <div id="message"></div>
    <script>
        let generatedOTP = "";
        function generateOTP() {
            // Generate 6-digit OTP using Math.random()
            generatedOTP = Math.floor(100000 + Math.random() * 900000);
            // For demo purposes (normally you wouldn't show this)
            alert("Generated OTP: " + generatedOTP);
            document.getElementById("message").innerHTML = "";
        }
        function verifyOTP() {
            let userInput = document.getElementById("userOTP").value;
            if (userInput.length !== 6) {
                document.getElementById("message").innerHTML = "Please enter a 6-digit
OTP";
                document.getElementById("message").style.color = "orange";
                return;
            }
            if (userInput === generatedOTP) {
                document.getElementById("message").innerHTML = "Valid OTP";
                document.getElementById("message").style.color = "green";
            } else {
                document.getElementById("message").innerHTML = "Invalid OTP";
                document.getElementById("message").style.color = "red";
            }
        }
    </script>
</body>
</html>

```

3. Write a program to create a simple Calculator with history.

Requirements:

- Perform +, −, ×, ÷
- Store last 5 calculations on the page

Use:

- #Array
- #DOM dynamic update

```
<!DOCTYPE html>
<html>
<head>
  <title>Calculator with History</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
      margin-top: 40px;
    }
    input {
      padding: 8px;
      width: 80px;
    }
    button {
      padding: 8px 12px;
      margin: 5px;
    }
    #history {
      margin-top: 20px;
    }
  </style>
</head>
<body>
  <div>
    <h2>Calculator</h2>
    <input type="number" id="num1">
    <br></br>
    <input type="number" id="num2">
  </div>
  <br><br>
  <button onclick="calculate('+')">+</button>
  <button onclick="calculate('-')">-</button>
  <button onclick="calculate('*')">*</button>
```

Calculator

+

-

*

/

Result: 6

Last 5 Calculations

4 * 5 = 20

6 + 5 = 11

60 / 5 = 12

60 - 54 = 6

```

<button onclick="calculate('/')">/</button>
<h3 id="result">Result: </h3>
<h3>Last 5 Calculations</h3>
<ul id="history"></ul>
<script>
  let historyList = [];
  function calculate(op) {
    let n1 = parseFloat(document.getElementById("num1").value);
    let n2 = parseFloat(document.getElementById("num2").value);
    let res;
    if (isNaN(n1) || isNaN(n2)) {
      alert("Enter numbers");
      return;
    }
    if (op === '+') res = n1 + n2;
    if (op === '-') res = n1 - n2;
    if (op === '*') res = n1 * n2;
    if (op === '/') res = n1 / n2;
    document.getElementById("result").textContent = "Result: " + res;
    let text = n1 + " " + op + " " + n2 + " = " + res;
    historyList.push(text);
    // keep only last 5
    if (historyList.length > 5) {
      historyList.shift();
    }
    displayHistory();
  }
  function displayHistory() {
    let ul = document.getElementById("history");
    ul.innerHTML = "";
    historyList.forEach(item => {
      let li = document.createElement("li");
      li.textContent = item;
      ul.appendChild(li);
    });
  }
</script>
</body>
</html>

```

4. Write a program to create a live search filter for a list of items.

Requirements:

- Display 10 items in list
- As user types, matching items remain visible

Use:

- #onkeyup event
- #filter()
- #DOM hide/show

```
<!DOCTYPE html>
<html>
<head>
  <title> Live search filter</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      padding: 20px;
    }
    input {
      margin: 5px;
      padding: 5px;
    }
    button {
      padding: 5px 10px;
      margin: 5px;
    }
    /* .student-item {
      margin: 8px 0;
      padding: 8px;
      background-color: #f4f4f4;
      border-radius: 5px;
    } */
  </style>
</head>
<body>
  <h2>Live search</h2>
  <input type="text" id="search" placeholder="Search item..."
onkeyup="filterStudents()">
  <div id="list">
  </div>
<script>
```

Live search

- Keyboard
- Mouse
- Laptop
- GPU
- CPU
- Headphone
- AirPods
- Mobile
- Charger
- Monitor

```

    const list = ["Keyboard", "Mouse", "Laptop", "GPU", "CPU", "Headphone",
"Airpods", "Mobile", "Charger", "Monitor"];
    function displayStudents(list) {
        const d = document.getElementById("list")
        let ul = document.createElement("ul");
        list.forEach((item) => {
            const li = document.createElement("li");
            li.textContent = item;
            ul.appendChild(li);
        })
        d.appendChild(ul);
    }
    function filterStudents() {
        let d = document.getElementById("list");
        d.removeChild(d.lastElementChild)
        const searchValue = document.getElementById("search").value.toLowerCase();
        const filtered = list.filter(function(item) {
            return item.toLowerCase().includes(searchValue);
        });
        displayStudents(filtered);
    }
    displayStudents(list);
</script>
</body>
</html>

```

5. Write a program to build a countdown timer.

Requirements:

- User enters seconds
- Timer counts down
- Show “Time Over” when finished

Use:

- #setInterval()
- #DOM update

```

<!DOCTYPE html>
<html>
<head>
    <title>Countdown Timer</title>
</head>
<body>

```

```

<h2>Countdown Timer</h2>
<input type="number" id="seconds" placeholder="Enter seconds">
<button onclick="startTimer()">Start</button>
<h3 id="display">0</h3>
<script>
    let timer;
    function startTimer() {
        let time = parseInt(document.getElementById("seconds").value);
        document.getElementById("seconds").textContent = "Enter seconds";
        if (isNaN(time) || time <= 0) {
            alert("Enter valid seconds");
            return;
        }
        clearInterval(timer);
        document.getElementById("display").textContent = time;
        timer = setInterval(function () {
            time--;
            document.getElementById("display").textContent = time;
            if (time <= 0) {
                clearInterval(timer);
                document.getElementById("display").textContent = "Time Over";
                setTimeout(function () {
                    document.getElementById("display").textContent = "";
                    document.getElementById("seconds").value = "";
                }, 2000);
            }
        }, 1000);
    }
</script>
</body>
</html>

```

Countdown Timer

0

Countdown Timer

25

Countdown Timer

10

6. Write a program to validate a full Registration Form with multiple rules.

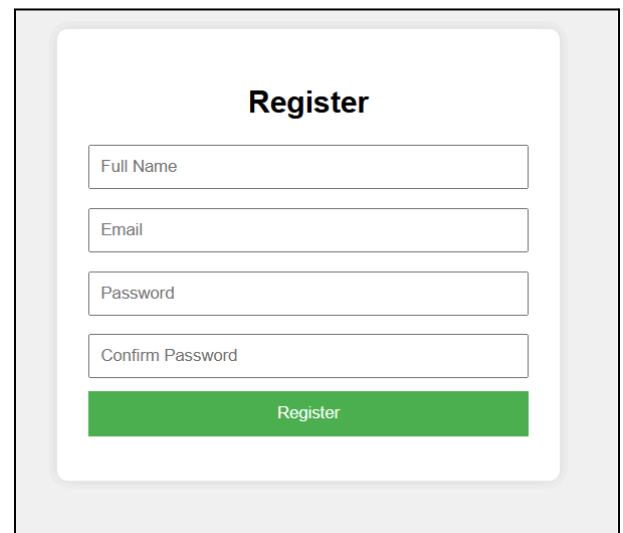
Rules:

- Name not empty
- Email format correct
- Password strong (uppercase + number)
- Confirm password match

Use:

- #Regex
- #Error messages below fields

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form Validation</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      padding: 40px;
    }
    .form-container {
      width: 350px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 8px;
      box-shadow: 0 0 10px rgba(0, 0, 0, 0.1);
    }
    h2 {
      text-align: center;
    }
    input {
      width: 100%;
      padding: 8px;
      margin-bottom: 10px;
      box-sizing: border-box;
    }
    button {
      width: 100%;
      padding: 10px;
      background: #4CAF50;
      color: white;
```



Register

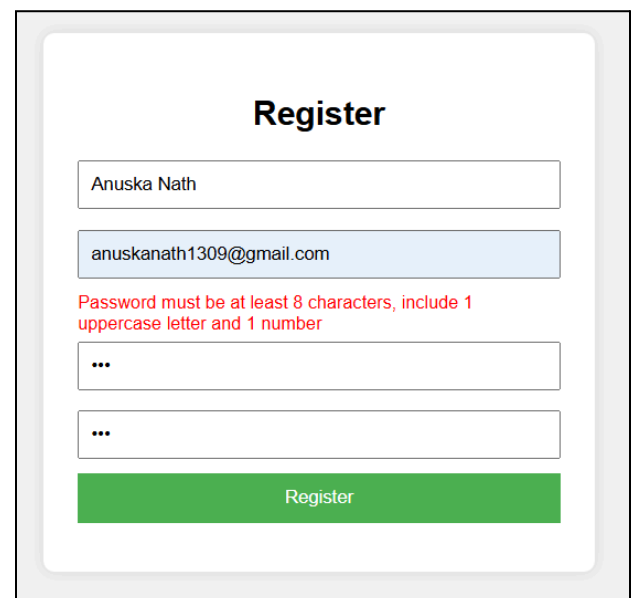
Full Name

Email

Password

Confirm Password

Register



Register

Anuska Nath

anuskanath1309@gmail.com

Password must be at least 8 characters, include 1 uppercase letter and 1 number

...

...

Register

```

        border: none;
        cursor: pointer;
    }
    button:hover {
        background: #45a049;
    }
    .error {
        color: red;
        font-size: 13px;
        margin-bottom: 5px;
    }
    .success {
        color: green;
        text-align: center;
        margin-top: 10px;
    }
</style>
</head>
<body>
    <div class="form-container">
        <h2>Register</h2>
        <form id="registerForm">
            <div id="nameError" class="error"></div>
            <input type="text" id="name" placeholder="Full Name">
            <div id="emailError" class="error"></div>
            <input type="text" id="email" placeholder="Email">
            <div id="passwordError" class="error"></div>
            <input type="password" id="password" placeholder="Password">
            <div id="confirmError" class="error"></div>
            <input type="password" id="confirmPassword" placeholder="Confirm
Password">
            <button type="submit">Register</button>
            <div id="successMessage" class="success"></div>
        </form>
    </div>
    <script>
        const form = document.getElementById("registerForm");
        form.addEventListener("submit", function(event) {
            event.preventDefault();
            document.getElementById("nameError").textContent = "";
            document.getElementById("emailError").textContent = "";
            document.getElementById("passwordError").textContent = "";

```

```

document.getElementById("confirmError").textContent = "";
document.getElementById("successMessage").textContent = "";
let isValid = true;
const name = document.getElementById("name").value.trim();
const email = document.getElementById("email").value.trim();
const password = document.getElementById("password").value;
const confirmPassword = document.getElementById("confirmPassword").value;
const emailRegex = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
const passwordRegex = /^(?=.*[A-Z])(?=.*\d){8,}$/;
if (name === "") {
    document.getElementById("nameError").textContent = "Name cannot be
empty";
    isValid = false;
}
if (!emailRegex.test(email)) {
    document.getElementById("emailError").textContent = "Enter a valid email
address";
    isValid = false;
}
if (!passwordRegex.test(password)) {
    document.getElementById("passwordError").textContent =
"Password must be at least 8 characters, include 1 uppercase letter and 1
number";
    isValid = false;
}
if (password !== confirmPassword) {
    document.getElementById("confirmError").textContent =
"Passwords do not match";
    isValid = false;
}
if (isValid) {
    document.getElementById("successMessage").textContent =
"Registration successful!";
}
});
</script>
</body>
</html>

```

7. Write a program to store and display user login session using LocalStorage.

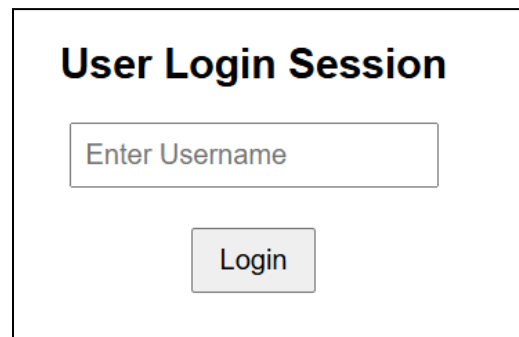
Requirements:

- Save username
- Show “Welcome back” after refresh
- Logout clears storage

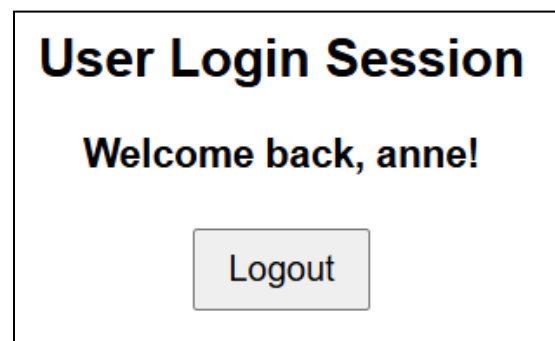
Use:

- #localStorage.setItem()
- #removeItem()

```
<!DOCTYPE html>
<html>
<head>
  <title>Login Session with LocalStorage</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 100px;
    }
    input {
      padding: 8px;
      font-size: 16px;
    }
    button {
      padding: 8px 15px;
      font-size: 16px;
      margin: 5px;
      cursor: pointer;
    }
    #message {
      margin-top: 20px;
      font-size: 18px;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <h2>User Login Session</h2>
  <div id="loginSection">
    <input type="text" id="username" placeholder="Enter Username">
    <br><br>
    <button onclick="login()">Login</button>
```



A rectangular box representing a login form. At the top, it has the title "User Login Session" in bold black text. Below the title is a text input field with the placeholder text "Enter Username". Underneath the input field is a button labeled "Login".



A rectangular box representing a login form after a successful login. At the top, it has the title "User Login Session" in bold black text. Below the title is a bold black message "Welcome back, anne!". Underneath the message is a button labeled "Logout".

```

</div>
<div id="welcomeSection" style="display:none;">
  <p id="message"></p>
  <button onclick="logout()">Logout</button>
</div>
<script>
  // Check if user already logged in (after refresh)
  window.onload = function() {
    let storedUser = localStorage.getItem("username");

    if (storedUser) {
      showWelcome(storedUser);
    }
  };
  function login() {
    let username = document.getElementById("username").value;

    if (username === "") {
      alert("Please enter username");
      return;
    }
    // Save username in localStorage
    localStorage.setItem("username", username);
    showWelcome(username);
  }
  function showWelcome(username) {
    document.getElementById("loginSection").style.display = "none";
    document.getElementById("welcomeSection").style.display = "block";
    document.getElementById("message").innerHTML = "Welcome back, " +
username + "!";
  }
  function logout() {
    // Remove username from localStorage
    localStorage.removeItem("username");
    document.getElementById("welcomeSection").style.display = "none";
    document.getElementById("loginSection").style.display = "block";
    document.getElementById("username").value = "";
  }
</script>
</body>
</html>

```

8. Write a program to create an Image Slider (Next/Previous).

Requirements:

- Display one image at a time
- Next/Prev buttons change image
- Auto-slide every 3 seconds

Use:

- #Array of images
- #setInterval()
- #DOM src update

```
<!DOCTYPE html>
<html>
<head>
  <title>Image Slider</title>
  <style>
    body {
      text-align: center;
      font-family: Arial, sans-serif;
      margin-top: 50px;
    }
    img {
      width: 400px;
      height: 250px;
      object-fit: cover;
      border-radius: 10px;
    }
    button {
      padding: 8px 15px;
      margin: 10px;
      font-size: 16px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h2>Image Slider</h2>
  <img id="slider" src="" alt="Slider Image">
  <br>
  <button onclick="prevImage()">Previous</button>
  <button onclick="nextImage()">Next</button>
  <script>
    // Array of images
```

```

const images = [
  "https://picsum.photos/id/1015/600/400",
  "https://picsum.photos/id/1016/600/400",
  "https://picsum.photos/id/1018/600/400",
  "https://picsum.photos/id/1020/600/400"
];
let currentIndex = 0;
const slider = document.getElementById("slider");
// Display first image
slider.src = images[currentIndex];
function showImage(index) {
  slider.src = images[index]; // DOM src update
}
function nextImage() {
  currentIndex = (currentIndex + 1) % images.length;
  showImage(currentIndex);
}
function prevImage() {
  currentIndex = (currentIndex - 1 + images.length) % images.length;
  showImage(currentIndex);
}
// Auto-slide every 3 seconds
setInterval(nextImage, 3000);
</script>
</body>
</html>

```

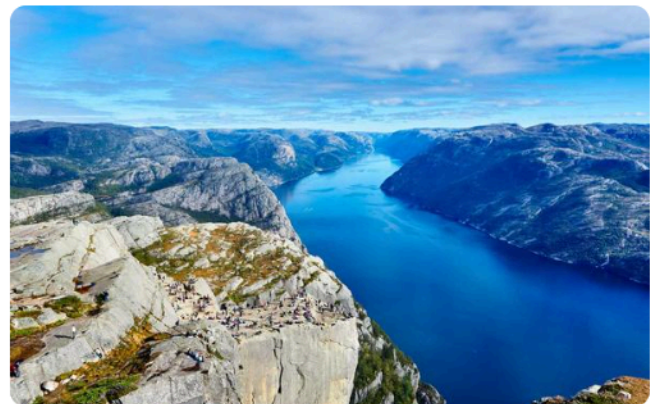
Image Slider



Previous

Next

Image Slider



Previous

Next

9. Write a program to create a Quiz Application.

Requirements:

- Display one question with options
- User selects answer
- Show score at end

Use:

- #Objects/Arrays
- #DOM dynamic content

```
<!DOCTYPE html>
<html>
<head>
  <title>Quiz Application</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin-top: 50px;
    }
    .option {
      display: block;
      margin: 8px;
    }
    button {
      padding: 8px 15px;
      margin-top: 15px;
      font-size: 16px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h2>Simple Quiz App</h2>
  <div id="quiz-container">
    <h3 id="question"></h3>
    <div id="options"></div>
    <button onclick="nextQuestion()">Next</button>
  </div>
  <div id="result" style="display:none;"></div>
<script>
  // Quiz questions (Array of Objects)
  const quizData = [{
```

Simple Quiz App

Which language is used for web structure?

☐ HTML

☐ CSS

☐ JavaScript

☐ Python

Next

Simple Quiz App

Your Score: 3 / 3


```

        question: "Which language is used for web structure?",
        options: ["HTML", "CSS", "JavaScript", "Python"],
        correct: 0
    }, {
        question: "Which method is used to select an element by ID?",
        options: ["querySelector()", "getElementById()", "getElementsByClass()",
"selectElement()"],
        correct: 1
    }, {
        question: "Which company developed JavaScript?",
        options: ["Microsoft", "Google", "Netscape", "Apple"],
        correct: 2
    }
    ]];
let currentQuestion = 0;
let score = 0;
const questionEl = document.getElementById("question");
const optionsEl = document.getElementById("options");
const resultEl = document.getElementById("result");
function loadQuestion() {
    const currentQuiz = quizData[currentQuestion];
    questionEl.innerText = currentQuiz.question;
    optionsEl.innerHTML = ""; // Clear previous options
    currentQuiz.options.forEach((option, index) => {
        optionsEl.innerHTML += `
            <label class="option">
                <input type="radio" name="answer" value="${index}">
                ${option}
            </label>
        `;
    });
}
function nextQuestion() {
    const selected = document.querySelector('input[name="answer"]:checked');
    if (!selected) {
        alert("Please select an answer!");
        return;
    }
    if (parseInt(selected.value) === quizData[currentQuestion].correct) {
        score++;
    }
    currentQuestion++;
    if (currentQuestion < quizData.length) {

```

```

        loadQuestion();
    } else {
        showResult();
    }
}
function showResult() {
    document.getElementById("quiz-container").style.display = "none";
    resultEl.style.display = "block";
    resultEl.innerHTML = `

### Your Score: ${score} / ${quizData.length}</h3>`; } // Load first question loadQuestion(); </script> </body> </html>


```

10. Write a program to fetch and display JSON data from a file using Fetch API.

Requirements:

- Load student JSON file
- Display in table
- Add search option

Use:

- #fetch()
- #JSON.parse()
- #DOM table

Students.json:

```

[
  { "id": 1, "name": "Amit Sharma", "age": 20, "course": "BCA" },
  { "id": 2, "name": "Priya Singh", "age": 22, "course": "BBA" },
  { "id": 3, "name": "Rahul Verma", "age": 21, "course": "B.Tech" },
  { "id": 4, "name": "Sneha Patel", "age": 23, "course": "MBA" }
]

```

Q10.html:

```

<!DOCTYPE html>
<html>
<head>
  <title>Fetch JSON Students</title>
  <style>
    body {

```

```

        font-family: Arial, sans-serif;
        text-align: center;
        margin-top: 40px;
    }
    table {
        margin: 20px auto;
        border-collapse: collapse;
        width: 70%;
    }
    th,
    td {
        border: 1px solid #333;
        padding: 8px;
    }
    th {
        background-color: #f2f2f2;
    }

    input {
        padding: 6px;
        width: 200px;
    }
</style>
</head>
<body>
    <h2>Student Data</h2>
    <input type="text" id="search" placeholder="Search by name..."
onkeyup="searchStudent()">
    <table>
        <thead>
            <tr>
                <th>ID</th>
                <th>Name</th>
                <th>Age</th>
                <th>Course</th>
            </tr>
        </thead>
        <tbody id="tableBody">
        </tbody>
    </table>
    <script>
        let students = [];

```

```

// Load JSON file using fetch()
fetch("students.json")
  .then(response => response.text())
  .then(data => {
    students = JSON.parse(data); // Convert JSON string to object
    displayStudents(students);
  })
  .catch(error => console.error("Error loading JSON:", error));
// Display data in table (DOM manipulation)
function displayStudents(data) {
  const tableBody = document.getElementById("tableBody");
  tableBody.innerHTML = "";
  data.forEach(student => {
    tableBody.innerHTML += `
      <tr>
        <td>${student.id}</td>
        <td>${student.name}</td>
        <td>${student.age}</td>
        <td>${student.course}</td>
      </tr>
    `;
  });
}
// Search function
function searchStudent() {
  const searchValue = document.getElementById("search").value.toLowerCase();
  const filtered = students.filter(student =>
    student.name.toLowerCase().includes(searchValue)
  );
  displayStudents(filtered);
}
</script>
</body>
</html>

```

Student Data			
Search by name...			
ID	Name	Age	Course
1	Amit Sharma	20	BCA
2	Priya Singh	22	BBA
3	Rahul Verma	21	B.Tech
4	Sneha Patel	23	MBA

Student Data			
Rahul			
ID	Name	Age	Course
3	Rahul Verma	21	B.Tech

11. Write a program to implement a Dark/Light Theme Toggle with saved preference.

Requirements:

- Toggle theme button
- Theme saved even after reload

Use:

- `#classList.toggle()`
- `#localStorage`

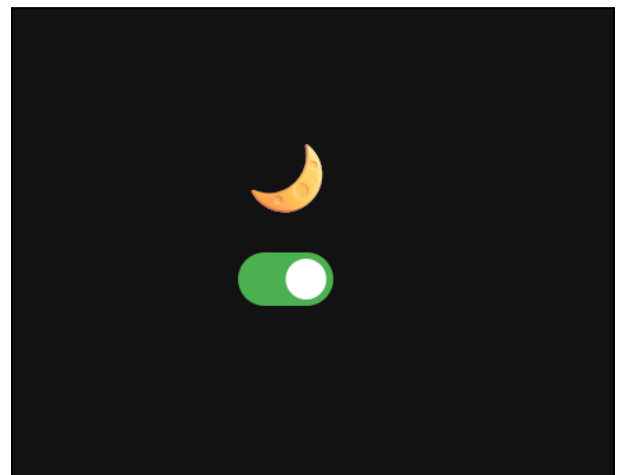
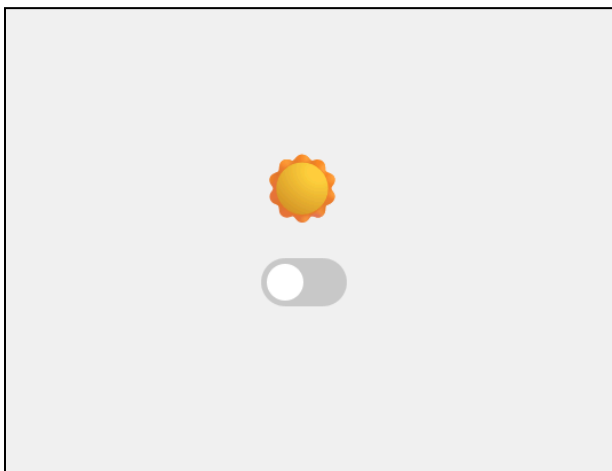
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1" />
  <title>Dark/Light Theme Toggle</title>
  <style>
    html,
    body {
      margin: 0;
      padding: 0;
      height: 100%;
      transition: background-color 0.3s, color 0.3s;
      font-family: Arial, sans-serif;
    }
    body.light {
      background-color: #f4f4f4;
      color: #222222;
    }
    body.dark {
      background-color: #121212;
      color: #ffffff;
    }
    .container {
      display: flex;
      flex-direction: column;
      justify-content: center;
      align-items: center;
      height: 100%;
      user-select: none;
    }
    .icon {
      font-size: 48px;
      margin-bottom: 20px;
    }
```

```
}
label.switch {
  position: relative;
  display: inline-block;
  width: 60px;
  height: 34px;
}
label.switch input {
  opacity: 0;
  width: 0;
  height: 0;
}
.slider {
  position: absolute;
  cursor: pointer;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background-color: #ccc;
  border-radius: 34px;
  transition: 0.4s;
}
.slider:before {
  position: absolute;
  content: "";
  height: 26px;
  width: 26px;
  left: 4px;
  bottom: 4px;
  background-color: white;
  border-radius: 50%;
  transition: 0.4s;
}
input:checked+.slider {
  background-color: #4caf50;
}
input:checked+.slider:before {
  transform: translateX(26px);
}
</style>
</head>
```

```

<body>
  <div class="container">
    <div class="icon" id="themeIcon">☀️</div>
    <label class="switch">
      <input type="checkbox" id="themeToggle" />
      <span class="slider"></span>
    </label>
  </div>
  <script>
    const body = document.body;
    const toggle = document.getElementById('themeToggle');
    const icon = document.getElementById('themeIcon');
    // Load saved theme or default to light
    let savedTheme = localStorage.getItem('theme') || 'light';
    // Apply saved theme class
    body.classList.add(savedTheme);
    toggle.checked = savedTheme === 'dark';
    icon.textContent = savedTheme === 'dark' ? '🌙' : '☀️';
    toggle.addEventListener('change', () => {
      if (toggle.checked) {
        body.classList.replace('light', 'dark');
        localStorage.setItem('theme', 'dark');
        icon.textContent = '🌙';
      } else {
        body.classList.replace('dark', 'light');
        localStorage.setItem('theme', 'light');
        icon.textContent = '☀️';
      }
    });
  </script>
</body>
</html>

```



12. Write a program to create a Real-Time Character Limit Checker.

Requirements:

- Textarea max 200 chars
- Show remaining characters
- Prevent typing after limit

Use:

- #input event
- #length validation

```
<!DOCTYPE html>
<html>
<head>
  <title>Character Limit Checker</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 50px;
      max-width: 500px;
    }
    textarea {
      width: 100%;
      height: 120px;
      font-size: 16px;
      padding: 10px;
      box-sizing: border-box;
      resize: vertical;
    }
    .remaining {
      margin-top: 5px;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <h2>Real-Time Character Limit Checker</h2>
  <textarea id="textInput" placeholder="Type here... (max 200
characters)"></textarea>
```



```

<div class="remaining" id="remainingCount">200 characters
remaining</div>
<script>
  const textarea = document.getElementById("textInput");
  const remaining = document.getElementById("remainingCount");
  const maxChars = 200;
  textarea.addEventListener("input", () => {
    let text = textarea.value;
    // If length exceeds max, trim the extra characters
    if (text.length > maxChars) {
      textarea.value = text.substring(0, maxChars);
      text = textarea.value;
    }
    // Update remaining characters
    const charsLeft = maxChars - text.length;
    remaining.textContent = `${charsLeft} character${charsLeft !== 1 ?
's' : ''} remaining`;
  });
</script>
</body>
</html>

```

Real-Time Character Limit Checker

Type here... (max 200 characters)

200 characters remaining

Real-Time Character Limit Checker

dhbxewnjqkamlskdbcfhvnftghfbnedwsma,mbfhjjnvmkcfrd3ew

147 characters remaining